

REVIT STEEL

Steel structural modeling in Autodesk Revit. The track case study: twenty nine modules from datum setup to fabrication detail.

29

MODULES IN FULL

101

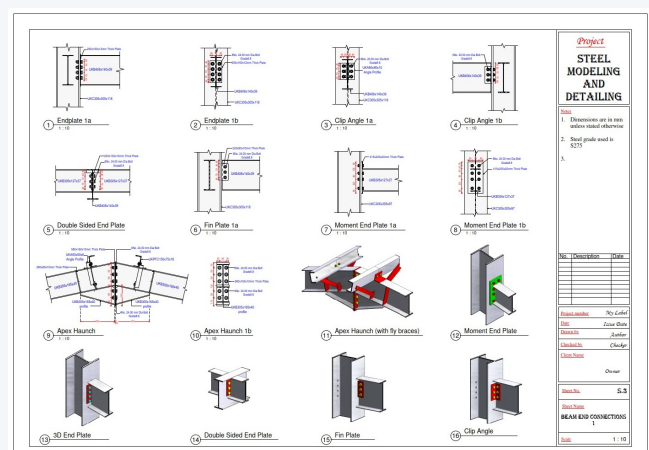
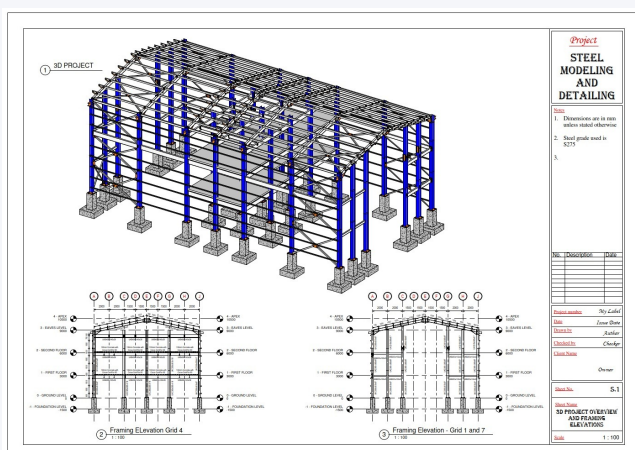
SUBTOPICS

4 wks

PER TRACK

100%

LIVE ONLINE



01 | CONTEXT

BACKGROUND

Background

Revit Steel Structure Training is a live instructor led track within the Structra Revit for Structures program. Over four weekday evening weeks you move from the Revit interface, levels and grids through steel and concrete columns, foundations, rafters, steel beams, purlins and bracing, then into connections, fabrication elements, schedules and sheets.

The track is 100% practical and built for engineering students and early career structural professionals who want to produce real, buildable steel models.

START DATE

Monday, 14
September 2026,
third week of
September

FORMAT

100% online, live
sessions led by an
instructor on Zoom

DURATION

4 weeks per track,
weekday evenings

FEE

KSH 2,500 per track

WHO IT'S FOR

University
engineering students
and early career
structural and BIM
professionals

PREREQUISITE

A laptop capable of
running Revit, no
prior Revit experience
required

02 | BEFORE TRAINING

THE CHALLENGE

The challenge

01

Who it is for

University engineering students and early career structural and BIM professionals who want to build real, buildable models, not just pretty renders.

02

Starting level

No prior Revit experience is required. The only prerequisite is a laptop capable of running Autodesk Revit.

03

Format and pace

100% online, live sessions led by an instructor on Zoom, weekday evenings, four weeks per track.

04

The gap it closes

Moving from knowing Revit exists to modeling steel structures end to end, with details, schedules and sheets.

03 | CURRICULUM

THE SOLUTION

The solution, module by module

All 29 modules in full, 101 practical subtopics, covered in the live sessions.

01 Brief overview of Revit (what it entails)

Difference with CAD software; Benefits over CAD software.

02 Setting up the software

Brief introduction of Libraries required; Downloading other extensions from Autodesk site.

03 Beginning with Revit

Creating a new project; An overview of the templates; An overview of palettes, tabs and panels.

04 Creating Datum Elements

Creating Levels; Creating Grids; Elevation Markers; Propagate options extents on the grid; Customizing and changing the shape and appearance of grid bubbles (optional); Linking CAD files to create grids.

05 Modeling Steel Columns

Placement of columns; Loading column families.

06 Modeling Concrete Columns

Placement of concrete columns; Loading column families.

07 Modeling the Foundations

Hosting foundations on the columns created; Loading other foundation families into your drawings.

08 Modeling the rafters

Placement of Rafters; Loading steel beam sections; Attaching Columns to the rafters; Copying rafters from grid to grid.

09 Modelling of the Steel Beams

Loading steel beam sections; Modelling of the floor beams; Justification Rule of the Beams; Copying of the floor beams; Modelling of the Beams at the Eaves Level.

10 Beam System Layout and Justification Rules

Using the beam system to model the structural beams; Modelling the beams using the beam system layout rules; Modelling the beams using the beam system Justification Rules; Deselecting the beam system.

11 Composite Slab and Composite Beam

Modelling the Composite Beam; Modelling the Composite Slab; Cantilever the Composite Slab.

12 Floor Openings

Using the Shaft Opening Tool.

13 Curved Beams

Procedure of modeling the curved beams.

14 Sloped/Slanted Beams

Modelling sloped beams using the offset tool; Modelling sloped beams using the Framing Elevation.

15 Purlins and Side Rails

Modelling the Purlins using Reference Planes; Modelling the Purlins using the Beam System; Modelling the Side Rails.

16 Bracings

Vertical Bracings; Horizontal Bracing.

17 Visibility/Graphics and Filters

Applying Visibility/Graphic Overrides , Method 1; Applying Visibility/Graphic Overrides , Method 2; Applying Visibility/Graphic Overrides , Method 3; Applying Visibility/Graphic Overrides , Method 4; Hiding Elements.

18 Creating Sections

Types of Sections; Creating sections in your drawing.

19 Placing of Reinforcement

Setting up the Reinforcement Cover; Placing Reinforcement at the Foundation; Column Reinforcement; Slab reinforcement.

20 Splitting Elements

Splitting Elements.

21 Introduction to Connections

Uploading the Connections; Applying Connections to the Structural Elements; Propagate the Connections.

22 Connection Types

Main Types of Connection; Haunch Connection; Apex Connection; Base Plate Connection; Gusset Plate Bracing Connection; Gusset Plate Column Base Plate Connection; Purlin Connection / Side Rail Connection; Single sided end plate (simple connection); Single sided end plate (moment connection); Double sided end plate (simple connection); Fin plate (simple connection); Clip Angle (simple connection); Column Splice connection; Creating Custom Connections.

23 Fabrication Elements, Modifiers and Parametric Cuts

Use and Location.

24 Annotations

Adding dimensions; Adding spot elevations; Adding spot coordinates; Adding spot slope; Creating Custom Bolt and Plate tags.

25 Detailing of Sections

Placing breaklines in the sections; Using hatch files to create regions; Placement of rebar tags on your reinforcements; Adding comments on your reinforcement; Tagging of the structural elements.

26 Schedules and Material Takeoffs

Creating Bar Bending Schedules; Loading shape images into bar bending schedules; Creating Material Takeoffs for the structural elements; Sorting and filtering fields in material takeoffs and bar bending schedules.

27 Placing Drawings on Sheets

Loading different paper sizes for your title blocks; Using system loaded title blocks; Editing and customizing the title blocks; Adjusting texts and fields in your title block; Saving title blocks as a Revit family template; Introduction to duplicate options for drawings; Placing drawings on the sheets.

28 Printing Sheets to PDF format

Setting up your sheets before printing; Printing one sheet at a time; Printing several sheets at a time.

29 Conversion of your files into CAD and other formats

Export options for your drawing into other formats; Transporting your drawing to AutoCAD; Sending your model for analysis.

04 | OUTCOME

THE RESULT

The result, what you will be able to build

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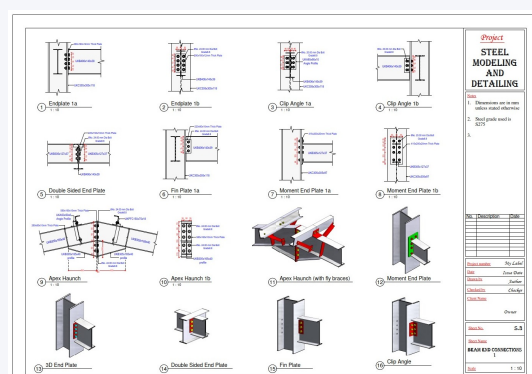
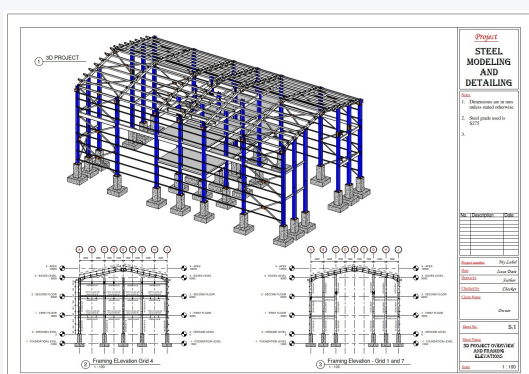
MODULES

101

SUBTOPICS

4 wks

DURATION



Learners complete the track able to model complete steel structures: rafters, floor beams, purlins and bracing, detailed down to haunch, apex, base plate and gusset plate connections, ready for fabrication data. The structures shown above are modeled during the program.

REVIT STEEL STRUCTURE TRAINING, AUTODESK REVIT

STRUCTRA

ENROL, NEXT COHORT 14 SEPT 2026

structratraining.co.ke · +254 7XX XXX XXX · Start 14 September 2026 · KSH 2,500 per track